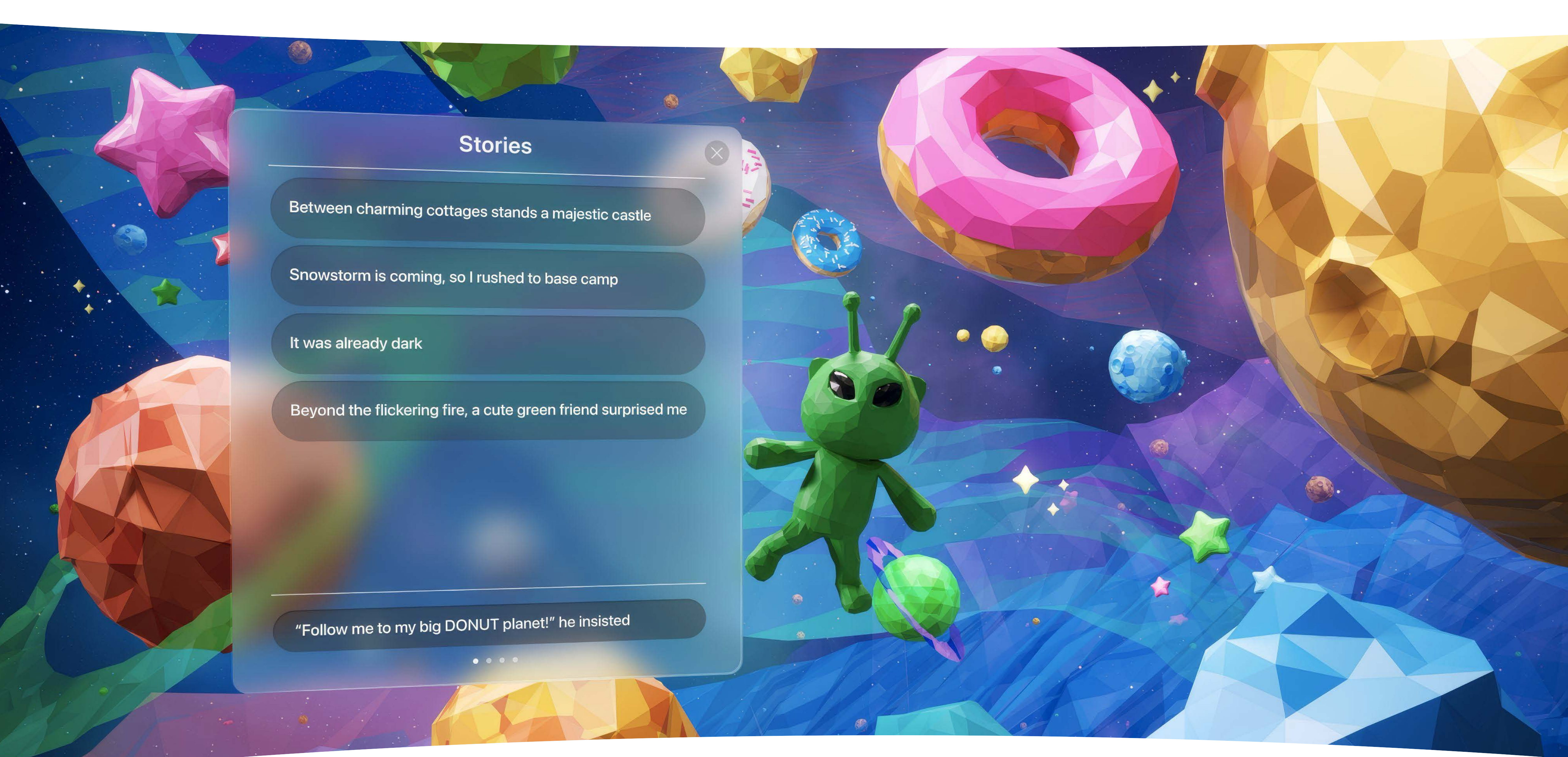




Stories

Between charming cottages stands a majestic castle

Snowstorm is coming, so I rushed to base camp

It was already dark

Beyond the flickering fire, a cute green friend surprised me

"Follow me to my big DONUT planet!" he insisted



Catalyzing Creativity with
AI-infused 3D Interactive Environment
Designed for Children's Story Writing

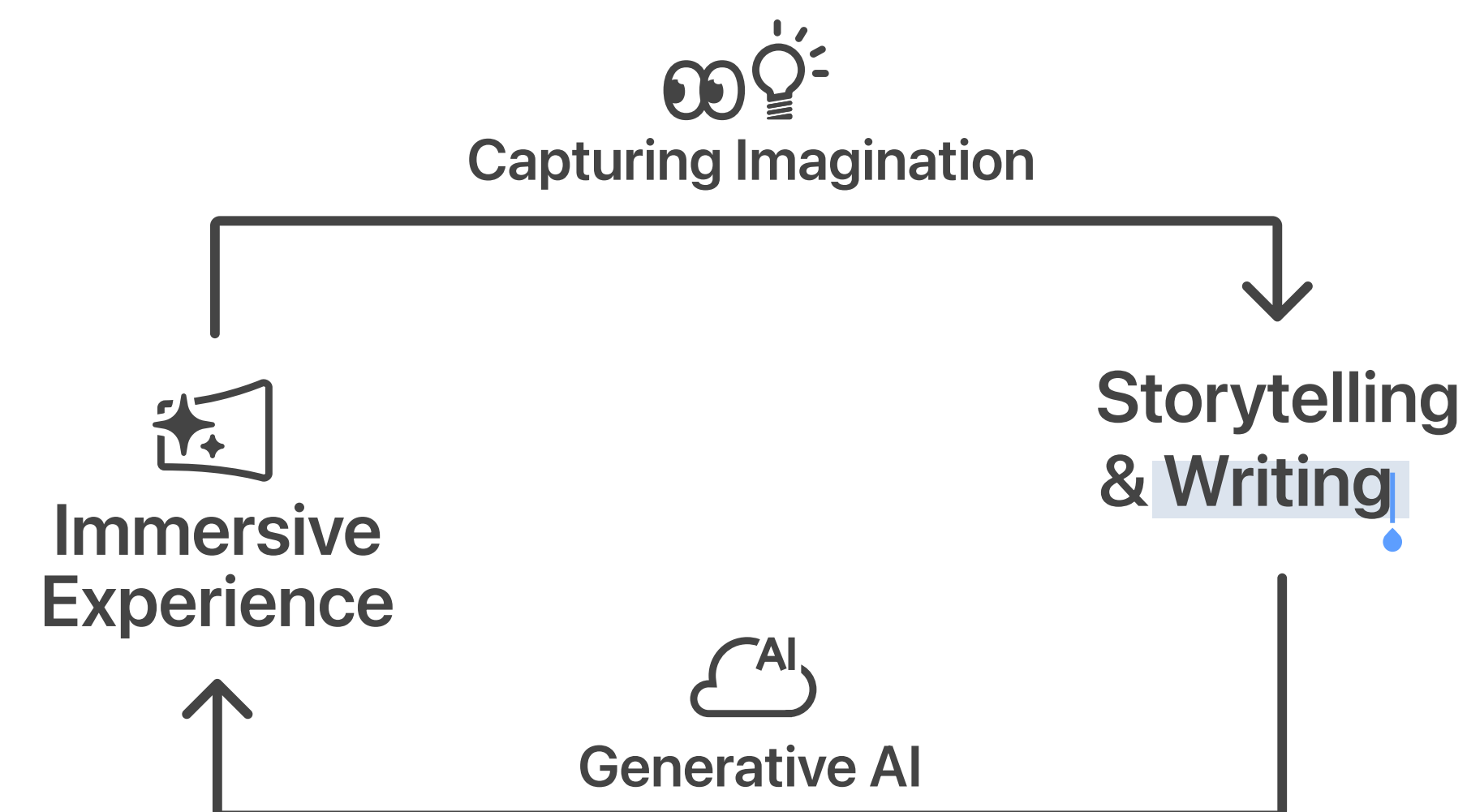


WrightHere : Project Overview

Cultivating a child's innate capacity for imaginative thinking and storytelling remains the cornerstone of creative development. Even with the rise of generative AI, many young writers still face writer's block, lacking the right catalyst to ignite their thoughts. WrightHere reimagines this challenge as an opportunity for exploration by transforming the abstract act of fictional imagination into an immersive experience. It is a 3D-AI platform where children can *dive into* their own story worlds through extended reality (XR), expand their ideas through interaction, and return to text with renewed inspiration.

Within every child lie countless untold stories—all they need is the confidence to begin and a companion to guide their journey. Moving beyond the conventional 'read more, write more' advice, WrightHere leverages human-AI teamplay, enabling AI to serve as a creative partner that guides, provokes, and co-creates. In this human-AI collaboration, WrightHere bridges the boundary between storytelling and world-building.

Technically, the platform integrates multi-modal generative AI (text, image, 3D, and audio) within an immersive XR environment, enabling children to interact with their narratives in real time. Grounded in *the Cognitive Process Theory of Writing*, WrightHere's dual-mode interface allows for seamless transitions between writing and 3D exploration, intuitively dismantling writer's block. Rather than following a linear progression from idea to story, the system encourages children to revisit, reinterpret, and reimagine their ideas—forming a continuous loop of inspiration and reflection. This circular design scaffolds natural creative jumps, sustaining engagement and imagination over time.



UX SYSTEM

WrightHere operates through two interconnected modes: Write Mode and Dive Mode. In Write Mode, users create stories in the (C) Narrative Editor, which automatically generates a corresponding 3D Story World. Clicking the (D) Story Thumbnail next to a paragraph transitions users to Dive Mode, where they can interact via (E) dialogue exchanges, (F) character/object manipulation, or (G) adding new 3D assets. These interactions are captured as (A) Story Materials, which can be dragged into (C) Narrative Editor for (B) AI-assisted co-writing function. The editor remains accessible in both modes, supporting a seamless, cyclical creative process.





Embodied Interaction

To extend WrightHere into immersive environments, we developed an XR version in Apple Vision Pro and conducted user testing with children. While short-term creative satisfaction showed modest changes, the XR modality significantly heightened users' sense of presence and involvement in storytelling.

Using the Igroup Presence Questionnaire (0–6 scale), participants transitioning from PC to XR reported notable increases in spatial presence (3.53→4.73, $p=.006$) and involvement (2.69→3.66, $p=.004$). These results highlight how XR can deepen narrative engagement, transforming writing into a lively embodied experience.



Design Considerations

Design Goal

Considerations

1. Creative Agency

The system should lower the initial hurdle when children get stuck, sustaining engagement.

Enable a generative environment that supports rapid iteration, utilizing a dual-mode interface to nudge interaction switching.

2. Contextual AI Alignment

AI must achieve alignment with the author's intent regarding the story progression.

Apply context engineering to maintain narrative coherence, while maximizing generative flexibility to integrate diverse user intents.

3. Modality & Immersion Design

The XR modality should facilitate a seamless transition from 2D screens to 3D immersion.

Leverage HMD-native gestures and gaze to facilitate spatial interaction, prioritizing immersive experiences over unnecessary flat UI.

Form

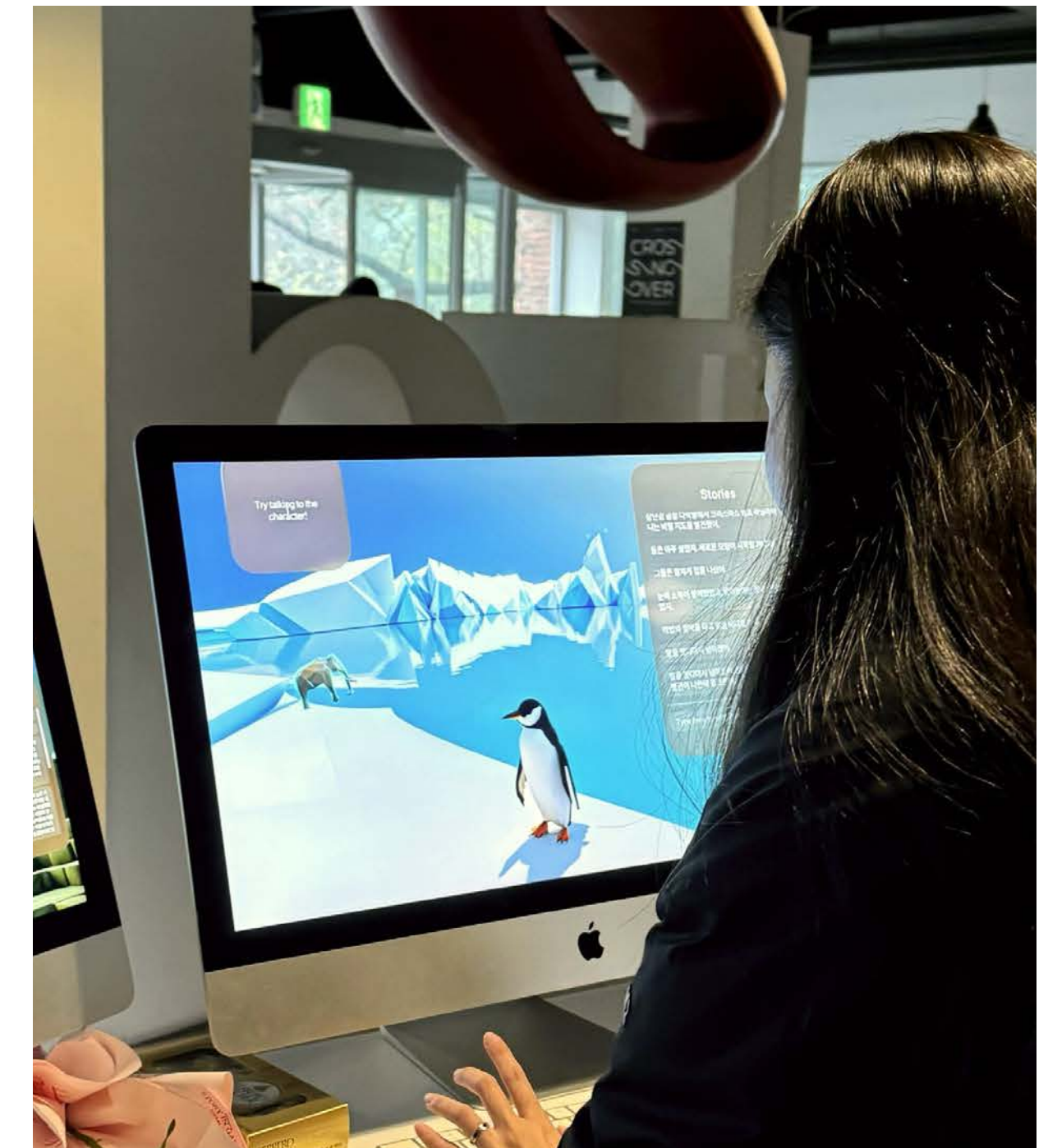
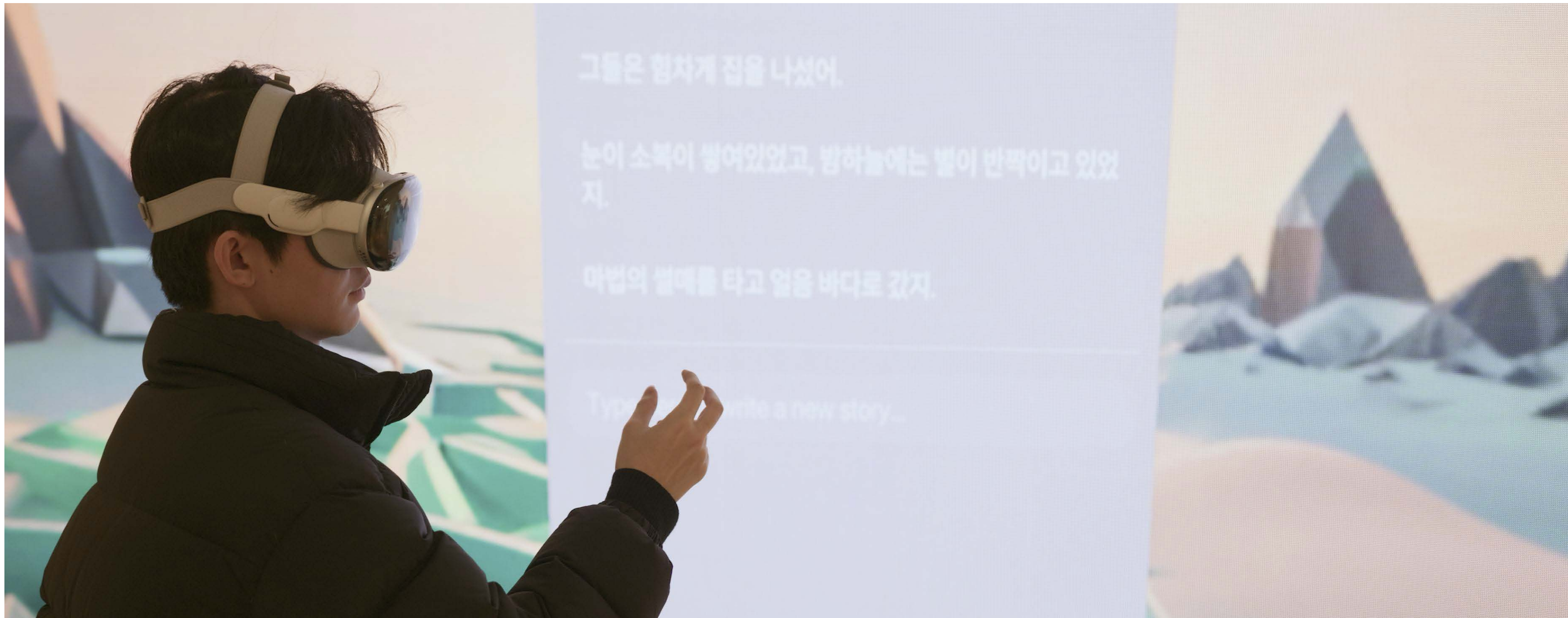
Designed for children aged 11–15, WrightHere adopts a low-polygon visual style rather than photorealism—a deliberate choice to preserve room for imagination and ensure age-appropriate visuals. Instead of over-defining visual details, the simplified aesthetic invites children to project their own ideas into the story world, fostering creativity through abstraction. Story spaces are rendered with 360-degree images to minimize latency, allowing rapid iteration. WrightHere fosters emotional engagement by allowing them to explore personalized 3D spaces and interact with characters that serve as tangible anchors for empathy and narrative immersion.

Function

WrightHere is designed for intuitive use: as users write, the platform continuously interprets their narrative, storing data to improve AI understanding. The system operates through a context-aware information architecture with a LLM at its core, calling generative AI APIs to dynamically update story content. The dual-mode design—Write Mode and Dive Mode—ensures seamless transitions between composing text and exploring the 3D narrative world. Interactions in Dive Mode are captured as ‘Story Materials’ that can be reused in Write Mode, enabling rapid iteration and creative continuity. Users can gain inspiration and develop storytelling skills.

Sustainability

In an age overwhelmed by content, the ability to critically engage with media and create one's own stories is more vital than ever. WrightHere lowers the barriers to storytelling by offering an accessible, creative space where children can turn imagination into narrative. Socially, it nurtures confidence, empathy, and critical thinking, inspiring future storytellers. Moreover, in a media ecosystem where users are often passive consumers reliant on providers, WrightHere empowers children to become creators themselves—transforming consumption into participation. This shift from reliance to authorship fosters a more sustainable content culture.



Different Usage Patterns Among Children

We conducted a two-phase user study with 32 participants using the WrightHere system to explore how children engage with dual-mode creative writing and how AI and immersive design support creativity. Participants freely wrote original stories of any topic or style, followed by surveys and interviews evaluating usability, presence, and creativity. Usage patterns varied widely: some participants focused on switching between Write and Dive modes, using AI as a co-writer or dialogue partner, while others concentrated on world-building—creating and placing 3D objects within the scenes. These diverse strategies reveal how children balanced exploration, AI collaboration, and imagination in shaping their creative process.



Use Case Scenario

Ethan, a 13-year-old writer, starts in the Narrative Editor drafting his story about Christmas elves. He enters Dive Mode, creates a sleigh object and an elf character named Tinsel, and asks “Can we fly?” Hearing “Yes, hop on!” from Tinsel, he explores the snowy scene, adjusts characters with gesture in XR HMD, and looks around to immerse himself. By returning to Write Mode, Ethan cycles between spatial exploration and story development.

Findings of a thematic analysis of qualitative data from a user study

Our study with children demonstrated that the system supports creative story writing by helping them overcome writer's block, sustain narrative momentum, and externalize imagination through multimodal interaction in 3D story world. The following elaborates on four major aspects of the thematic analysis from the user study.

Curiosity as a Narrative Driver

Children's ability to sustain narrative drive was closely tied to curiosity, which became the underlying force that kept them both enjoying and progressing the system. Rather than passively receiving output, children actively questioned characters and environments—about what to do next, where they were, or who a character might be—and these inquiries propelled the story forward.

Aligned Surprise and Serendipity

Children came to expect AI to provide aligned yet surprising contributions—clarifying their latent thoughts and visualizing the unimaginable. This sense of anticipation redefined AI not as a mere source of inspiration, but as a thought-provoking collaborator capable of triggering new imaginative possibilities.

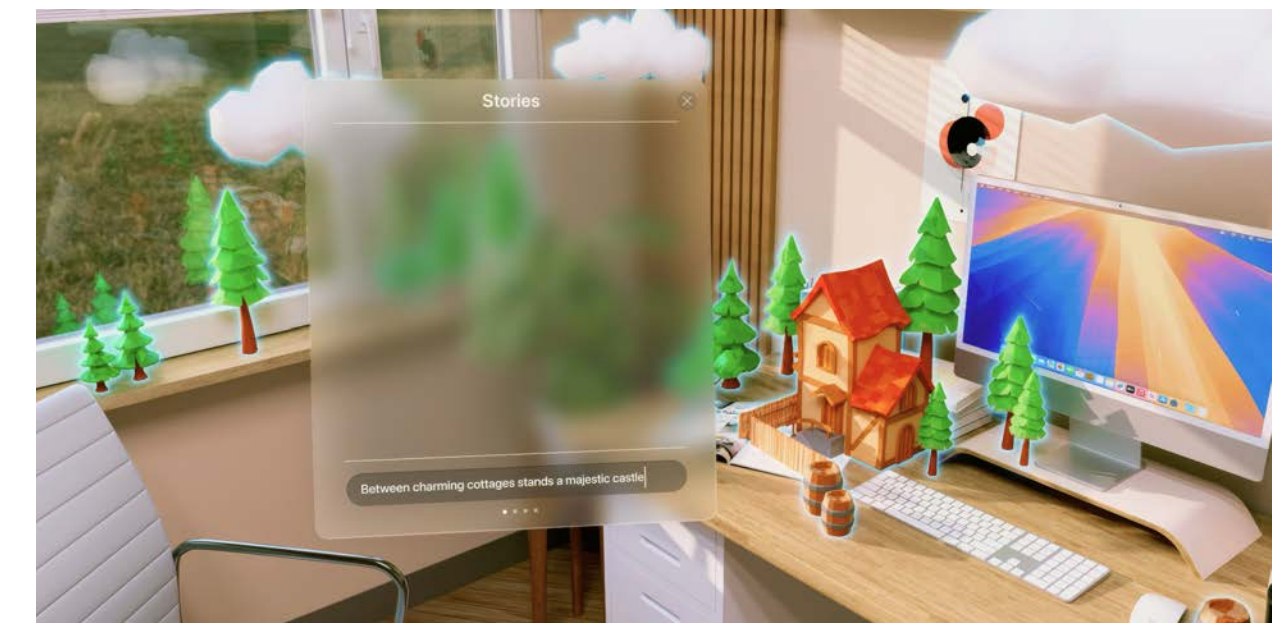
Becoming the Hero of the Story

Participants emphasized how Dive Mode enabled them to step directly into their narratives as protagonists. Rather than imagining scenarios from a distance, they described an embodied sense of 'being there' that transformed their engagement. One child explained: "It felt like I actually became the main character and went on the adventure. It was like I went inside a book." Similarly, another child noted, "Because I could feel everything I wrote right away, it was easier to immerse myself."

Creating Effects Through Spatial Manipulation

Beyond building static scenes, children used XR's affordances—embodied gesture and movements—to generate effects that shaped narrative flow. Manipulation became a way to influence atmosphere and pacing, as one participant noted: "I made the spider suddenly grow bigger to give a jump scare." One another participant emphasized embodied interaction: "Unlike on PC, here I could put objects right in front of me, and it felt more vivid." By leveraging embodied control within immersive space, children treated manipulation itself—whether in size, speed, or proximity—as a way to shape pacing, mood, and causality in their stories.

“The system helps children ask not only ‘What happens next?’ but also ‘Why?’, supporting deeper reflection in storytelling education.”



CURRENT STAGE

WrightHere has launched its Proof-of-Concept in PC environment and MVP in the extended reality environment of Apple Vision Pro. A user study with 50+ children/adults has been conducted to assess creativity, usability, and engagement. The findings are being integrated into the next iteration of service launch in 2026, refining both the AI collaboration model and the interface for wider deployment.